Chapter 25

Pathway Enrichment and Molecular Network Analysis: Over-Representation Analysis, GSEA, and Cross-Omics Topological Networks

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Abstract

High-resolution mass spectrometry experiments frequently culminate in ranked lists of hundreds of differentially abundant peptides, proteins, or metabolites. However, an uncurated feature list does not provide mechanistic clarity: cellular phenotype is mediated not by isolated molecules, but by coordinated fluxes through metabolic cascades, multi-protein complexes, and signaling cascades. Bridging physical mass spectra to biological insight requires functional enrichment and molecular network analysis.

This chapter establishes the statistical, topological, and computational foundations for pathway-level inference in proteomics and metabolomics. We develop the mathematical framework of Over-Representation Analysis (ORA) via the hypergeometric distribution, Fisher's exact test, and non-central Wallenius formulations, deriving an analytical proof demonstrating why using the whole genome rather than the empirically detected mass spectrometry background creates massive false-positive inflation. We formalize functional class scoring via Gene Set Enrichment Analysis (GSEA), deriving the weighted Kolmogorov-Smirnov running-sum statistic, normalized enrichment score (NES), and permutation testing. We then transition to graph-theoretic network biology, formalizing adjacency matrices, degree and betweenness centrality, random walk with restart (RWR), Signaling Pathway Impact Analysis (SPIA), and Louvain community detection. Finally, we formulate bipartite cross-omics network integration between proteomics and metabolomics, supported by a tutor diagnostic heuristics checklist, an end-to-end reproducible R laboratory, a review of the "Five Deadly Sins" of omics pathway analysis, and multi-tier exercises with complete worked solutions.

Keywords: Pathway Enrichment, Over-Representation Analysis (ORA), GSEA, Hypergeometric Test, Background Universe Bias, Network Topology, Betweenness Centrality, Bipartite Graphs, Multi-Omics Integration, SPIA.

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25.1 Tutor's Methodological Roadmap for Pathway & Molecular Network Analysis

Before executing pathway enrichment packages or loading protein interaction graphs, the researcher must identify the analytical scope and appropriate statistical model. Table 25.1 summarizes the four sequential checkpoints required to extract publication-grade functional insights.

Table 25.1: Tutor's Methodological Roadmap: Four Checkpoints for Pathway & Network Biology

Stage	Physical Mechanism	Key Analytical Objective	Primary Failure Mode & Remedy
1. Functional Paradigm	Threshold-based selection (ORA) vs. continuous ranking (GSEA) vs. topological signaling (SPIA) [24, 15, 26].	Select method aligned with biological signal: discrete hits vs. coordinated sub-threshold shifts.	Using ORA when changes are subtle and coordinated. Remedy: Deploy fast preranked GSEA (`fgsea`) [17].
2. Universe Curation	MS detection bias toward high-abundance housekeeping enzymes [29].	Restrict ORA background universe strictly to the set of quantified proteins (N_{MS}).	Whole-genome reference ($N \approx 20,000$) creates massive false-positive inflation. Remedy: Use quantified MS background [30].
3. Network Topology	Physical protein-protein interaction (PPI) complexes and signaling cascades [2, 25].	Quantify node degree, betweenness centrality, and Louvain modularity clusters [4].	Hairball visualization without modular clustering. Remedy: Filter high-confidence edges (> 0.70); apply Louvain community detection [27].
4. Multi-Omics Graphs	Bipartite enzyme-metabolite biochemical reaction networks [22, 28].	Map protein enzymes and metabolic substrates into integrated bipartite graphs.	Unconnected multi-omics lists. Remedy: Query KEGG/Reactome reaction databases [14, 11].

Methodological Note

Tutor's Methodological Advice: The Background Universe Axiom
Never test a list of differentially abundant mass spectrometry proteins against the complete human genome ($N \approx 20,000$). High-abundance housekeeping pathways (e.g., Ribosome, Proteasome, TCA Cycle) represent 3% of the mass spectrometry proteome, but only 0.4% of the genome. Testing against the whole genome inflates the apparent odds ratio by nearly 8-fold, guaranteeing false-positive pathway discoveries out of pure technical sampling bias [29, 30].

25.2 Introduction: From Molecular Catalogs to Mechanistic Topology

The output of a rigorous differential abundance analysis (Chapters 21–22) is a list of features accompanied by effect sizes ($\log_2$FC), test statistics (t), and adjusted false discovery rates (q-values) [23, 20]. While essential, this list presents three major interpretation dilemmas:

1. **High-Dimensional Redundancy:** Biologically related proteins in a multi-subunit complex (e.g., the 26S proteasome, respiratory chain complexes I–V) exhibit correlated abundance shifts. Treating them as independent discoveries over-counts biological signals.

2. **Modest-Effect Synergy:** Metabolic pathways often respond through subtle, coordinated shifts (e.g., 20–30% abundance changes across 15 pathway enzymes) that fail individual univariate significance thresholds ($\alpha = 0.05$ after Benjamini-Hochberg correction [3]), yet collectively shift metabolic flux.

3. **Uncurated Heterogeneity:** Without functional grouping, discovering 18 kinase subunits, 12 ribosomal proteins, and 5 heat shock chaperones appears disconnected, obscuring underlying cellular stress or translational arrest.

To extract biological mechanism from mass spectrometry catalogs, computational omics relies on two complementary paradigms: **Enrichment Analysis** (evaluating whether predefined gene sets are statistically over-represented or coordinate along a continuum [24, 19]) and **Network Topology** (mapping detected features onto physical protein-protein interaction [PPI] graphs or metabolic reaction networks to identify regulatory bottlenecks and functional modules [2, 25]).

25.3 Over-Representation Analysis and the Background Universe Bias

25.3.1 Hypergeometric and Fisher's Exact Formulation

Over-Representation Analysis (ORA) evaluates whether a selected subset of significant features (e.g., proteins with $q < 0.05$ and $|\log_2 \text{FC}| > 1.0$) contains more members of a canonical pathway $\mathscr{P}$ than expected by random chance.

Let:

- N denote the total number of features in the reference *background universe*.
- M denote the number of features in the background universe belonging to pathway $\mathscr{P}$.
- n denote the number of differentially abundant (significant) features selected by the analyst ($n \leq N$).
- k denote the number of significant features that belong to pathway $\mathscr{P}$ ($k \leq \min(n, M)$).

This configuration corresponds to sampling without replacement from a finite population, represented by the 2×2 contingency table:

Table 25.2: Contingency table for Over-Representation Analysis of pathway $\mathscr{P}$.

	In Pathway $\mathscr{P}$	**Not in Pathway $\mathscr{P}$**	**Total**
Differentially Abundant	k	$n-k$	n
Non-Significant	$M-k$	$(N-M)-(n-k)$	$N-n$
Total Background	M	$N-M$	N

Under the null hypothesis H_0 that differential abundance is independent of pathway membership, the test statistic K follows the hypergeometric distribution:

$$P(K = k) = \frac{\binom{M}{k}\binom{N-M}{n-k}}{\binom{N}{n}} \tag{25.1}$$

The one-tailed p-value quantifying the probability of observing at least k pathway members by chance is:

$$p_{\text{ORA}} = P(K \geq k) = \sum_{i=k}^{\min(n,M)} \frac{\binom{M}{i}\binom{N-M}{n-i}}{\binom{N}{n}} \tag{25.2}$$

Eq. (25.2) corresponds exactly to the one-sided Fisher's Exact Test.

25.3.2 Proof: The Background Universe Selection Fallacy

Takeaway Summary

Theorem 25.1 (Background Universe Bias):
Let $N_{\text{genome}} \approx 20,000$ denote the total human protein-coding genome, and let $N_{\text{MS}} \approx 3,000$ denote the set of proteins actually detected and quantified in a mass spectrometry experiment ($N_{\text{MS}} \subset N_{\text{genome}}$). Assume an abundant pathway $\mathscr{P}$ (e.g., ribosome, glycolysis, TCA cycle) has M_{genome} total members, of which M_{MS} are detectable in MS ($M_{\text{MS}} \leq M_{\text{genome}}$). Because MS preferentially detects high-abundance housekeeping proteins, the baseline pathway density is substantially higher in the MS universe than in the genome:

$$\theta_{\text{MS}} = \frac{M_{\text{MS}}}{N_{\text{MS}}} > \frac{M_{\text{genome}}}{N_{\text{genome}}} = \theta_{\text{genome}} \tag{25.3}$$

If an analyst tests n significant MS hits against the whole genome background N_{genome}, the expected number of false-positive pathway overlaps under the null is deflated, resulting in artificially anti-conservative p-values:

$$p_{\text{genome}} \ll p_{\text{correct}} \tag{25.4}$$

Proof:
Under the whole-genome reference, the null expectation of overlap is:

$$\mathbb{E}[K \mid \text{Genome}] = n \cdot \frac{M_{\text{genome}}}{N_{\text{genome}}} = n\theta_{\text{genome}}$$

Under the correct MS reference universe, the null expectation is:

$$\mathbb{E}[K \mid \text{MS}] = n \cdot \frac{M_{\text{MS}}}{N_{\text{MS}}} = n\theta_{\text{MS}}$$

Since $\theta_{\text{MS}} > \theta_{\text{genome}}$, we have $\mathbb{E}[K \mid \text{MS}] > \mathbb{E}[K \mid \text{Genome}]$. When evaluating the tail probability $P(K \geq k)$, substituting N_{genome} places k much further into the upper tail of the artificially shifted null distribution:

$$\frac{k - n\theta_{\text{genome}}}{\sqrt{n\theta_{\text{genome}}(1-\theta_{\text{genome}})}} \gg \frac{k - n\theta_{\text{MS}}}{\sqrt{n\theta_{\text{MS}}(1-\theta_{\text{MS}})}}$$

Consequently, $p_{\text{genome}} \ll p_{\text{correct}}$. ■

25.3.3 Signaling Pathway Impact Analysis (SPIA)

Traditional ORA and GSEA treat pathways as mere bags of genes, ignoring biochemical wiring. Tarca and colleagues [26] formulated Signaling Pathway Impact Analysis (SPIA). SPIA combines standard enrichment evidence (P_{NDE}, probability of number of differentially expressed genes) with a perturbation accumulation factor (P_{PERT}) that models gene-gene interactions along directed signaling topologies (activation, inhibition, phosphorylation) via a system of linear equations:

$$\mathbf{a} = \mathbf{B}\beta + \mathbf{B}\mathbf{a} \implies \mathbf{a} = (\mathbf{I} - \mathbf{B})^{-1}\mathbf{B}\beta \tag{25.5}$$

where $\mathbf{B}$ is the adjacency matrix weighted by interaction type and β is the vector of measured log-fold changes. The global pathway significance combines both dimensions: $P_G = c - c\ln(c)$ where $c = P_{\text{NDE}} \cdot P_{\text{PERT}}$ [26].

25.4 Gene Set Enrichment Analysis: Functional Class Scoring

25.4.1 Mathematical Formulation of GSEA

Gene Set Enrichment Analysis (GSEA; Subramanian et al. [24]) bypasses thresholds by ranking all N quantified features along a continuous spectrum of differential abundance:

$$L = \{g_{(1)}, g_{(2)}, \ldots, g_{(N)}\}, \quad \text{where } s(g_{(1)}) \geq s(g_{(2)}) \geq \cdots \geq s(g_{(N)}) \tag{25.6}$$

The score $s(g)$ is typically the moderated t-statistic or signed log-fold change $\text{sign}(\log_2 \text{FC}) \times (-\log_{10} p)$.

GSEA evaluates whether features in gene set $\mathcal{S}$ are randomly distributed throughout L or enriched at the extremes. It computes a running-sum statistic by walking down the ranked list from index $i = 1$ to N:

$$P_{\text{hit}}(i, \mathcal{S}) = \sum_{j \leq i, g_{(j)} \in \mathcal{S}} \frac{|s(g_{(j)})|^p}{N_R}, \quad \text{where } N_R = \sum_{g \in \mathcal{S}} |s(g)|^p \tag{25.7}$$

$$P_{\text{miss}}(i, \mathcal{S}) = \sum_{j \leq i, g_{(j)} \notin \mathcal{S}} \frac{1}{N - N_H}, \quad \text{where } N_H = |\mathcal{S}| \tag{25.8}$$

The **Enrichment Score (ES)** is the maximum deviation of $P_{\text{hit}} - P_{\text{miss}}$ from zero:

$$\text{ES}(\mathcal{S}) = \max_{1 \leq i \leq N} \left(P_{\text{hit}}(i, \mathcal{S}) - P_{\text{miss}}(i, \mathcal{S}) \right) \tag{25.9}$$

To compare pathways of varying sizes, the score is normalized by the mean expected score under sample or gene permutations: $\text{NES} = \text{ES} / \mathbb{E}[\text{ES}_{\text{perm}}]$ [24, 17].

25.5 Network Topology and Graph-Theoretic Metrics

25.5.1 Adjacency Matrix, Centrality, and Modularity

Let $G = (V, E)$ denote an undirected molecular interaction network with vertex set V ($|V| = N$ proteins or metabolites) and edge set E ($|E| = M$ physical interactions) [9, 21].

- **Degree Centrality:** $k_i = \sum_{j=1}^{N} A_{ij}$, quantifying local connectivity.
- **Betweenness Centrality:** $C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$, where σ_{st} is the number of shortest paths between nodes s and t, and $\sigma_{st}(v)$ is the number of those paths passing through v. Proteins with high betweenness act as signaling bottlenecks [2].

- **Louvain Community Modularity:** Blondel et al. [4] and Traag et al. [27] partition networks into densely connected modules by maximizing modularity:

$$Q = \frac{1}{2M} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2M} \right] \delta(c_i, c_j) \tag{25.10}$$

where c_i is the module assignment of node i.

25.5.2 Random Walk with Restart (RWR)

To prioritize disease genes or identify sub-networks seeded by top mass spectrometry hits, Köhler et al. [16] formulated Random Walk with Restart. An imaginary walker traverses the network, transitioning with probability $(1-r)\mathbf{W}$ and restarting at seed nodes with probability $r\mathbf{p}_0$:

$$\mathbf{p}_{t+1} = (1-r)\mathbf{W}^T \mathbf{p}_t + r\mathbf{p}_0 \tag{25.11}$$

At steady state, the stationary probability vector $\mathbf{p}_\infty$ ranks all network nodes by functional proximity to the experimental seed proteins.

25.6 Hands-On R Laboratory: Pathway and Network Pipeline

R Lab Listing 1: Over-Representation Analysis with Background Universe Control [R 4.6 / Bioconductor]

```r
suppressPackageStartupMessages({
  library(dplyr)
  library(igraph)
  library(ggplot2)
})

# Construct synthetic background universe: 3,000 quantified MS proteins
set.seed(42)
ms_universe <- paste0("PROT_", sprintf("%04d", 1:3000))

# Define canonical pathways
pathway_db <- list(
  Glycolysis_TCA         = paste0("PROT_", sprintf("%04d", 1:80)),
  Inflammatory_Signaling = paste0("PROT_", sprintf("%04d", 81:120)),
  Olfactory_Receptors    = paste0("PROT_", sprintf("%04d", 121:125))
)

# Simulate 60 differentially abundant proteins (true enrichment in Glycolysis)
true_hits <- c(
  paste0("PROT_", sprintf("%04d", 1:25)),   # 25 Glycolysis hits
  paste0("PROT_", sprintf("%04d", 81:95)),  # 15 Inflammatory hits
  sample(ms_universe[126:3000], size = 20)  # 20 background random hits
)

# Custom ORA Function with Empirical Universe Control
audit_pathway_enrichment <- function(sig_genes, pathway_list, background_universe) {
  N <- length(background_universe)
  n <- length(sig_genes)
  res_list <- list()

  for (pname in names(pathway_list)) {
```

```
    p_genes <- intersect(pathway_list[[pname]], background_universe)
    M <- length(p_genes)
    overlap_genes <- intersect(sig_genes, p_genes)
    k <- length(overlap_genes)

    # One-sided Fisher / Hypergeometric test: P(X >= k)
    p_val <- phyper(k - 1, M, N - M, n, lower.tail = FALSE)
    odds_ratio <- ((k + 0.5) / (n - k + 0.5)) / ((M - k + 0.5) / (N - M - n + k + 0.5))

    res_list[[pname]] <- data.frame(
      Pathway       = pname,
      Universe_Size = M,
      Hit_Count     = k,
      Expected_Hits = round(n * (M / N), 2),
      Odds_Ratio    = round(odds_ratio, 2),
      P_Value       = p_val
    )
  }

  out_df <- do.call(rbind, res_list)
  out_df$FDR <- p.adjust(out_df$P_Value, method = "BH")
  return(out_df[order(out_df$P_Value), ])
}

test_res <- audit_pathway_enrichment(true_hits, pathway_db, ms_universe)
cat(sprintf("Audited %d pathways against MS universe of %d proteins.\n",
            nrow(test_res), length(ms_universe)))
print(test_res[, c("Pathway", "Universe_Size", "Hit_Count", "Expected_Hits", "P_Value",
    "FDR")])
```

```
Audited 3 pathways against MS universe of 3000 proteins. Pathway Universe_Size Hit_Count Expected_Hits P_Value FDR
1 Glycolysis_TCA 80 25 1.60 3.125419e-18 9.376257e-18 2 Inflammatory_Signaling 40 15 0.80 1.185240e-11 1.777860e-11
3 Olfactory_Receptors 5 0 0.10 1.000000e+00 1.000000e+00
```

25.7 The Five Deadly Sins of Omics Pathway Analysis

Methodological Pitfall & Deadly Sin

Deadly Sin 1: The Whole-Genome Background Fallacy.
Mechanism: Testing mass spectrometry hits against all 20,000 human genes rather than the 3,000 proteins detected on the instrument.
Consequence: Inflates odds ratios for housekeeping pathways by 5–10 fold, manufacturing dozens of false-positive pathways [29, 30].
Remedy: Strictly specify the background universe as all quantified features in the current experiment.

Methodological Pitfall & Deadly Sin

Deadly Sin 2: Threshold Cherry-Picking in Over-Representation Analysis.
Mechanism: Iteratively tweaking p-value and fold-change cutoffs until a preferred pathway passes significance ($p < 0.05$).

Consequence: Multiplicity of unadjusted searches invalidates false discovery rate guarantees.
Remedy: Use Gene Set Enrichment Analysis (GSEA) or pre-register significance thresholds [24, 15].

Methodological Pitfall & Deadly Sin

Deadly Sin 3: Disregarding Gene Set Overlap and Redundancy.
Mechanism: Reporting 40 significant Gene Ontology terms without accounting for hierarchical parent-child nesting.
Consequence: One single biological event (e.g., cell division) dominates results under 30 slightly differing term descriptions.
Remedy: Apply tree-simplification or semantic similarity clustering (e.g., `clusterProfiler::simplify`) [30].

Methodological Pitfall & Deadly Sin

Deadly Sin 4: Unweighted Protein Interaction Hairball Visualization.
Mechanism: Loading 500 proteins into Cytoscape, drawing all STRING interactions with confidence > 0.15, and presenting an unreadable dense blob.
Consequence: Fails to convey biological insight; obscures hub bottlenecks.
Remedy: Filter high-confidence interactions (> 0.70); apply Louvain community modularity or betweenness filtering [4, 2].

Methodological Pitfall & Deadly Sin

Deadly Sin 5: Omitting Topology in Cascading Signaling Networks.
Mechanism: Treating an essential receptor kinase identically to a peripheral transcription factor.
Consequence: Misses regulatory bottleneck shifts that direct downstream cascades.
Remedy: Deploy Signaling Pathway Impact Analysis (SPIA) or network flow propagation [26, 16].

25.8 Tutor's Self-Assessment Diagnostic Checklist

To evaluate the mathematical validity of your pathway enrichment and network analysis prior to publication, complete the diagnostic audit summarized in Table 25.3.

25.9 Chapter Exercises

Exercise 25.1 (Hypergeometric ORA Hand Calculation):
A mass spectrometry experiment quantifies $N = 4{,}000$ proteins. A metabolic pathway contains $M = 100$ proteins in this universe. The analyst selects $n = 80$ significant proteins, of which $k = 8$ belong to the pathway.

1. Compute the expected number of pathway hits under the null hypothesis $\mathbb{E}[K]$.
2. Compute the exact probability $P(K = 8)$ using Eq. (25.1).
3. Compute the sample odds ratio of enrichment.

Table 25.3: Tutor's Self-Assessment Diagnostic Checklist for Pathway Analysis

Step	Diagnostic Metric	Acceptable Threshold	Corrective Protocol
1	Background Universe Audit	N_{universe} equals total quantified MS proteins.	Eliminate whole-genome reference to prevent housekeeping bias [29].
2	Enrichment Strategy	GSEA applied when effect sizes are continuous or subtle.	Use fast preranked GSEA (`fgsea`) with $\geq$ 10,000 permutations [17].
3	Network Confidence Filtering	PPI interaction confidence score ≥ 0.700 (high confidence).	Discard low-confidence STRING edges to prevent hairball topology [25].
4	Modularity Partitioning	Louvain community modularity score $Q \geq 0.40$.	Cluster dense sub-networks before functional annotation [4].
5	Multiple Testing Control	Positive False Discovery Rate (pFDR) controlled via BH method.	Enforce pathway-level FDR ≤ 0.05 across all tested terms [3].

Exercise 25.2 (Graph Centrality and Bottleneck Identification):
Consider a 4-node star network where central node 1 is connected to three peripheral nodes 2, 3, and 4 (edges: $(1,2)$, $(1,3)$, and $(1,4)$).

1. Write the 4×4 adjacency matrix $\mathbf{A}$.
2. Compute the degree centrality of each node.
3. Compute the betweenness centrality of central node 1 versus peripheral nodes.

25.10 Comprehensive Worked-Out Solutions

Solution to Exercise 25.1: Hypergeometric Hand Calculation

1. Expected Overlap Under Null:

$$\mathbb{E}[K] = n \cdot \frac{M}{N} = 80 \cdot \frac{100}{4000} = 80 \cdot 0.025 = 2.0 \text{ proteins}$$

The observed overlap ($k = 8$) is $4\times$ higher than the null expectation.

2. Exact Hypergeometric Probability $P(K = 8)$:
From Eq. (25.1):

$$P(K = 8) = \frac{\binom{100}{8}\binom{3900}{72}}{\binom{4000}{80}}$$

Using R's `dhyper(8, 100, 3900, 80)`, we obtain:

$$P(K = 8) \approx 0.00114 \quad (0.114\%)$$

Evaluating the upper tail probability $P(K \geq 8)$ via `phyper(7, 100, 3900, 80, lower.tail = FALSE)`:

$$p_{\text{ORA}} = P(K \geq 8) \approx 0.00142$$

The pathway is statistically significantly over-represented ($p = 0.0014$).

3. Odds Ratio Calculation:
Contingency table entries:

$$k = 8, \quad n-k = 72, \quad M-k = 92, \quad N-M-n+k = 4000-100-72 = 3828$$

$$\text{OR} = \frac{k \cdot (N-M-n+k)}{(n-k)(M-k)} = \frac{8 \times 3828}{72 \times 92} = \frac{30624}{6624} \approx 4.623$$

Features in this pathway have a 4.62-fold higher odds of being differentially abundant than features outside the pathway [24, 15].

Solution to Exercise 25.2: Star Network Centrality

1. Adjacency Matrix:

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

2. Degree Centrality:

$$k_1 = 3, \quad k_2 = 1, \quad k_3 = 1, \quad k_4 = 1$$

3. Betweenness Centrality:
The pairs of non-adjacent nodes are: $(2,3),(2,4),(3,4)$ (3 pairs total). Every shortest path between these pairs must traverse central node 1 (e.g., $2 \to 1 \to 3$). Therefore, for node 1:

$$\sigma_{st}(1) = \sigma_{st} = 1 \implies C_B(1) = 3$$

Normalized by $\frac{(n-1)(n-2)}{2} = \frac{3\times 2}{2} = 3$:

$$C_B^*(1) = \frac{3}{3} = 1.0 \quad (100\% \text{ betweenness})$$

For peripheral nodes $v \in \{2,3,4\}$, no shortest path between any other pair passes through them, so $C_B(v) = 0$. Node 1 is a critical network hub and single point of regulatory failure [2].

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